Linear Algebra Project Instructions

Objective

This project connects the theory and computation of cubic spline interpolation. In Question 1, you proved that the tridiagonal systems arising from different spline boundary conditions are always well-posed, since their determinants are strictly positive for any $n \geq 3$ data points. In Question 2, you will apply this theory to real atmospheric CO_2 data: implementing natural and parabolic runout cubic splines, using them to estimate CO_2 concentrations for 1990 and 2010, and comparing your interpolated values against the actual historical data. The goal is to understand both why spline interpolation works (Question 1) and its practical strengths and limitations, especially when extrapolating beyond the data range (Question 2).

Question 1
For $n \geq 3$, let A_n, B_n and C_n be the following $n \times n$ matrices:

$$A_n = \begin{pmatrix} 4 & 1 & 0 & \dots & 0 & 0 & 0 \\ 1 & 4 & 1 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 4 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 & 4 \end{pmatrix}, \qquad B_n = \begin{pmatrix} 5 & 1 & 0 & \dots & 0 & 0 & 0 \\ 1 & 4 & 1 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 4 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 & 5 \end{pmatrix},$$

$$C_n = \begin{pmatrix} 6 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1 & 4 & 1 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 4 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 & 6 \end{pmatrix}.$$

Show that for $n \geq 3$, the determinants of A_n, B_n, C_n are all positive.

Question 2: Cubic Spline Interpolation
The estimated mean atmospheric concentration of carbon dioxide in earth's atmosphere is given in the following table. (`https://data.giss.nasa.gov/modelforce/ghgases/Fig1A.ext.txt`)

Year	CO_2 (ppm)
1850	285.2
1875	288.6
1900	295.7
1925	305.3
1950	311.3
1975	331.36
2000	369.64

1. Write a computer code to compute and plot the natural cubic spline through the data, and compute the spline estimate for the CO_2 concentration in 1990 and 2010. Compare you results with the real data in the above link.

2. Carry out the same analysis for the parabolic runout spline and the Cubic Runout Spline.

Project Requirements

1. Submit a detailed written report, including your source code (well-commented) as an appendix.
2. Include theoretical explanations (proofs for Question 1).
3. Use clear mathematical notation, and include plots/diagrams of the spline interpolations where necessary.
4. Clearly state and discuss your numerical results, including the spline estimates for 1990 and 2010, and compare them with the actual data values.

Deadline

The project deadline is Thursday, 16 July 2026, at 11:00 pm.